



HAPPY



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	7.95
2024	B.Tech in Artificial Intelligence and Data Science	IIITDM Kurnool	8.42
2020	High School / CBSE	JP DAV Public School, Ganaur	92.80
2018	Intermediate / CBSE	JP DAV Public School, Ganaur	87.00

IIT DELHI THESIS

Title: Representation Learning using ECG+PPG

Supervisor: Prof. Rahul Garg

Description: Developing a **multi-modal representation learning** framework to fuse **ECG and PPG** signals, creating a robust feature space to mitigate noise and artifacts in PPG data. This research focuses on developing a novel deep learning model that combines ECG and PPG signals to improve diagnostic accuracy. The project included clinical validation with data from **AIIMS Hospital**, demonstrating practical application.

INTERSHIPS

- **CSIR(Council of Scientific & Industrial Research) Research Internship , Chennai** (June, 2023 - July, 2023)
 - Developed a highly efficient **GAN model** to deal with the problem of limited training images which ultimately helps in developing a highly accurate Classification and Segmentation model with accuracy 1.00 in classification and IoU 0.70 in segmentation.
- **Samsung Prism Project Internship , Samsung R&D Institute (Virtual)** (Dec, 2022 - June, 2023)
 - Project involves the integration of UNet and GAN models, incorporating **Semi-supervision** and **Active Learning** techniques which improved **semantic segmentation** model accuracy from 0.66 to 0.73.
 - Successfully managed 3 complex datasets (Cityscapes, DLRSD, Minifrance) and collaborated closely with SamsungR&D group, aligning AI model outcomes with practical needs, showcasing adaptability and teamwork.

PROJECTS

- **Stroke Type Detection using Fundus Imaging (Minor Project) (Prof. Rahul Garg):**
 - Developed a deep learning model for stroke type detection using fundus imaging, fine-tuning the **RETFound foundation model** in a collaborative research project with AIIMS Delhi,
 - It focuses on applying computer vision to medical diagnostics. The project also involved multi-class classification for Diabetic Retinopathy and stroke detection.
- **LaTeX to Markdown Compiler (Prof. Smruti Ranjan Sarangi) :**
 - Tokenized LaTeX commands and structures using **Flex**. Parsed tokens into an Abstract Syntax Tree (AST) with **Bison** for accurate syntax recognition.
 - Developed a custom **AST traversal** mechanism to translate LaTeX syntax into equivalent Markdown representation.
- **Run and Gun Game in Unity (Prof. Smruti Ranjan Sarangi) :**
 - Designed multiple enemy types with unique behaviors and attack patterns. Integrated various power-ups, enhancing player abilities and gameplay variety. Implemented smooth player controls enabling actions like crouching, shooting, jumping, and power-up activation. Implemented parallax background, dynamic platforms, and multi-level environments for enhanced game interactivity.
- **Mini Web Browser (Prof. Smruti Ranjan Sarangi) :**
 - Engineered a mini web browser, implementing a robust fetching mechanism with libcurl and building a custom **HTML parser** using Flex and Bison to construct a DOM-based Abstract Syntax Tree (AST).
 - Developed a **multi-threaded** rendering engine with **Qt** to visualize the **HTML DOM**, supporting a multi-tab user interface with smooth UI updates and responsive performance.

TECHNICAL SKILLS

- **Languages:** C/C++, Python, Javascript, HTML, CSS, Flex, Bison, SQL
- **Libraries / Tools:** Tensorflow, Pytorch, Tkinter, Swing API(Java), Qt, VScode, Git, Github, Colab, Verilog, Unity

SCHOLASTIC ACHIEVEMENTS

- **Departmental Rank 6:** Current rank in M.Tech (Computer Science & Engineering), IIT Delhi.
- **Rank 1 in National Abacus Championship:** Got Rank-1 out of 2000 participants in withAbacus category and Third award in withoutAbacus category.

QUALIFYING EXAMS

- **Graduate Aptitude Test in Engineering (GATE) Rank: 29**



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IIT COURSE			
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M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	7.95	---
COURSES DONE			

Advanced Data Structures, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Teaching Practicum In Computer Science And Engineering, Computer Vision, Machine Learning, Minor Project, Teaching Practicum In Computer Science And Engineering